

Credit: BAE Systems



A battleground of the future

autonomous technologies used by many armies around the world, including the US, UK and Australia. Some of these are used for combat simulation purposes, to train soldiers for different scenarios, while others have actually been used in armed conflicts around the world. The Ironclad is a UGV developed for maneuvering in the tight spaces of urban conflict, but can just as easily handle cross country terrain. The single vehicle has a number of modules for operating in different scenarios. A remote turret allows soldiers to fire at enemies from a distance, it can double up as a bomb disposal robot, scout ahead in hazardous areas before soldiers enter the space, and team up with another Ironclad unit with a stretcher between the two, to evacuate wounded soldiers from the battlefield. These versatile robots are meant to be used in tandem with other UGVs, armoured vehicles and tanks on the battlefield, can move at convoy speeds, and can be deployed from other vehicles. This falls in line with the vision of BAE systems of how future battles will play out, with a number of land vehicles and aerial platforms sharing situational awareness and teaming up to track friendlies as well as enemies, with each unit making rapid decisions autonomously.

The US is not the only country in the world that is developing

autonomous tanks that can fight alongside conventional tanks as well as other ground units, UGVs and drones. Both Russia and China are developing such machines too, with Russia deploying the semi autonomous Uran-9 in the Syrian Civil War. Apparently the tanks did not perform as well as expected with their sensors not being up to the mark, and the communications being blocked by the surrounding buildings. However, the tanks were part of the large-scale military exercise Vostok 2018, in which China also participated, which was the first time a foreign power took part in such exercises. One of the capabilities that are being developed into the Uran-9 tanks is a spoken word interface that allows personnel to give verbal commands to the tanks. The Russians have also developed the autonomous Udar tanks that are configurable for a number of purposes, including reconnaissance and surveillance. The tank has been designed to navigate through rough terrains autonomously, and can work in tandem with tethered drones that can extend the surveillance range and capabilities. The Prohod-1 tanks can provide safe corridors for both personnel and cargo, and have been designed

to be operable in both manned and unmanned configurations. China is working on a number of different platforms, with sparse news coming out of the country, leading to much speculation of the capabilities of devices that have no name and are only known by blurry photos. Many of these tanks are being built to work in tandem with ground forces and other armoured units. China is also taking the innovative approach of upgrading old Russian tanks with autonomous capabilities. These tanks can be remotely controlled by an operator, but have the potential to be upgraded with even more autonomy. China is mostly using these tanks as realistic targets for anti tank missiles and artillery exercises.

A number of land vehicles, mostly tanks and armoured vehicles use what are known as hard kill countermeasures, which use some kind of scanning to disrupt incoming projectiles when detected, with explosives, rockets or machine gun fire. The Russian Arena system, the Israeli Trophy and Iron Fist systems, the



Credit: Lockheed Martin

The first truly modern ALV

American Quick Kill, Raven and Iron Curtain systems all use radio, radar, laser or infrared sensors to detect incoming projectiles, and return fire with kinetic countermeasures of their own. Some warships also use such autonomous systems to neutralise incoming cruise missiles, bombs and even artillery attacks. These are autonomous weapons that trigger when a threat is detected, and are used chiefly because the human reaction time is not sufficient to neutralise the incoming projectiles. ■